Intelligent University Observatory: Research and Lab Management System using Multi-Agent System

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1 Project Overview

The goal of this project is to develop a **Multi-Agent System (MAS)** to monitor, manage, and analyze researchers, laboratories, and research units within a university. The system aims to provide decision-makers with metrics on research activity, clustering of researchers, recommendations for collaboration, and insights using AI and optional LLM-based semantic analysis. The system will include:

- Metrics on researchers (active researchers, publications, citations, h-index, collaborations)
- Clustering and classification of researchers or labs
- Expertise and potential collaboration identification
- Optional LLM agents to detect similar publications and expertise areas
- Structured storage in a database
- Web-based interface and dashboard for administrators

2 Learning Objectives

- Design and implement a MAS with specialized agents
- Apply Game Theory and Negotiation techniques
- Integrate AI for clustering researchers and publications
- Utilize LLM agents for semantic analysis (optional)
- Develop a database schema for research data
- Create a web dashboard for visualization and decision-making

3 Proposed Agent Architecture

The proposed Multi-Agent System (MAS) architecture is composed of several specialized agents that collaboratively collect, analyse, and exploit research information in order to support collaboration discovery. The main components of the system are summarized as follows:

- **AgentCoordinator:** Orchestrates the overall system workflow by coordinating the activities of the different agents and managing the communication between data collection, analysis, and recommendation modules.
- **Observer Agents (AgentResearcherScraper, AgentPublicationScraper, AgentLabScraper):** These agents are responsible for collecting data from multiple online sources such as academic profiles, publication databases, and laboratory websites using web scraping techniques or APIs.

- **AgentCluster and AgentExpertiseMatcher:** These agents analyse the collected data by clustering researchers according to their expertise and identifying potential complementarities between research profiles.
- **AgentCollabAdvisor and AgentNegotiator:** Based on the analysis results, these agents recommend potential collaborations and simulate negotiation strategies to facilitate the formation of research partnerships.
- **AgentDashboardInterface:** Provides a visualization and interaction layer for managers and researchers, allowing them to explore suggested collaborations and monitor the system outputs.

4 Database Design

Suggested relational database schema:

Table	**Attributes**	**Description**
Researchers	researcher_id, name, lab_id, department, h_index, publications, citations	Researcher profile and metrics
Labs	lab_id, name, department, number_of_researchers, active_projects, University, Country	Laboratory data
Publications	publication_id, title, year, citations, authors	Publications metadata
Clusters	cluster_id, cluster_name, members	Clusters of researchers or labs
Expertise	researcher_id, expertise_area, keywords	Expertise mapping

You can refine this schema according to your project.

5 Python Libraries and Tools

- **MAS:** Mesa
- **Data Collection:** BeautifulSoup, Selenium, Requests
- **Data Analysis:** Pandas, NumPy, scikit-learn
- **Clustering/ML:** scikit-learn, TensorFlow, PyTorch (optional)
- **LLM Semantic Analysis (optional):** OpenAI API, HuggingFace Transformers
- **Web/Dashboard:** Flask, Django, Plotly Dash, Streamlit
- **Database:** SQLite, PostgreSQL, MongoDB

6 Expected Deliverables

1. Functional MAS with observer, profiling, recommendation/negotiation agents
2. Database with researcher, lab, and publication data
3. Web interface and interactive dashboard
4. Project report with architecture, agent descriptions, and workflows
5. Presentation and demo of the system

7 Project Planning (Semester)

Week	Phase	Description
1	Project Setup and Design	Team formation, requirements analysis, identification of data sources, database schema design, and definition of the multi-agent architecture.
2	Core Development	Implementation of observer/scraper agents for data collection and development of clustering and expertise profiling agents.
3	Advanced Development	Implementation of recommendation and negotiation agents and initial development of the web dashboard for visualization.
4	Integration and Finalization	Integration of MAS components with the database and web interface, system testing, debugging, and preparation of the final report and presentation.

8 Evaluation Criteria

- Functionality of MAS and agent interactions
- Correct implementation of clustering and AI agents
- Database design and integration quality
- Web dashboard usability and visualization
- Innovation in negotiation/game theory simulations
- Quality of documentation, report, and presentation

9 Optional Extensions

- Real-time monitoring of publications or researcher metrics
- Interactive visual analytics for collaboration decisions